

SmartHome Plate v2

REDESIGNED: 30% Cheaper, Same Durability

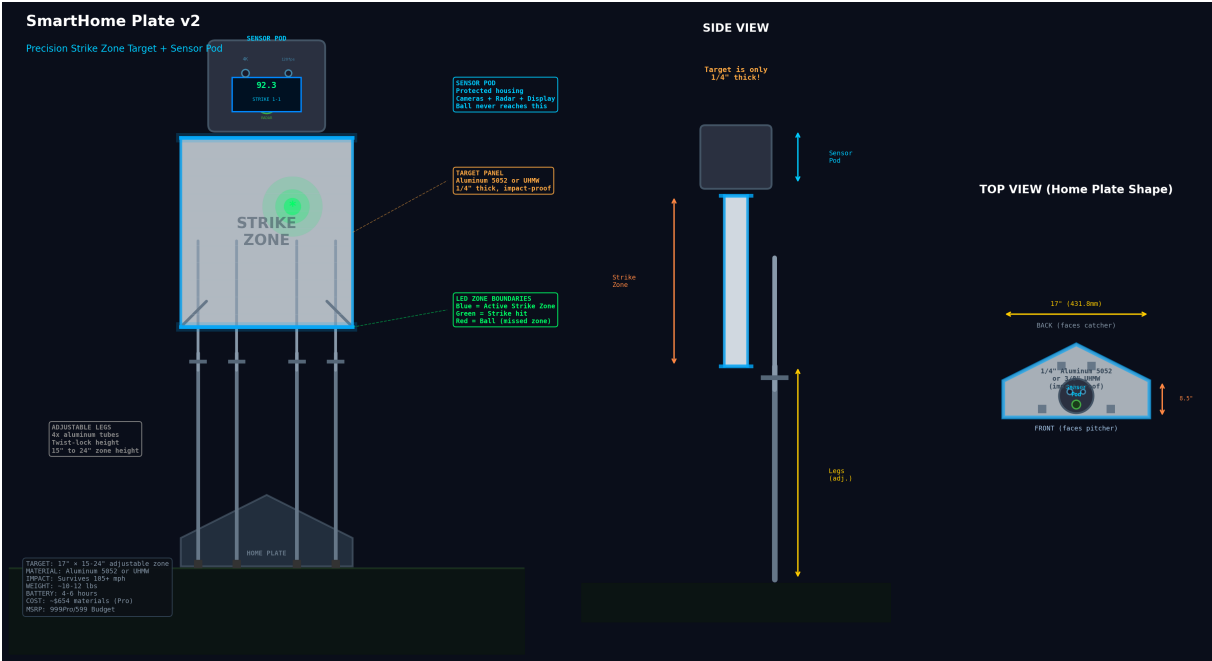
Strike Zone Target + Sensor Pod Architecture

Aluminum/UHMW Construction | 10-12 lbs | \$599-\$999 MSRP

Complete Build Guide with Cost Analysis

Pitching + Hitting Analytics | Spin Rate | Exit Velocity | Auto Zones

Version 2.0 - May 2025



1. v1 vs v2: Why the Redesign?

The v1 design was a 24" tall clear box sitting on home plate. It worked, but had fundamental problems:

- The entire 24" height was "target" - but the real strike zone is only 15-24" tall.
- 5 full polycarbonate panels = expensive (the biggest cost driver).
- Clear panels are cool-looking but add cost without adding durability.
- The box shape is heavy (20 lbs) and expensive to ship.
- Below the zone = wasted material. Above the zone = just protected sensors.

The v2 Insight: The Zone IS the Product

Instead of a big box, v2 is a precision strike zone TARGET PANEL on a lightweight frame. Only the zone area detects impacts. The sensor pod sits safely above. This cuts material cost by 30% while being MORE durable (aluminum survives anything a baseball can throw at it).

Side-by-Side Comparison

Aspect	v1 (Box)	v2 (Target)	Impact
Target surface	5 full panels, 24" tall	1 thin panel, zone-only	Less material
Frame	CNC aluminum plates	Aluminum tube frame	Cheaper mfg
Target material	3/8" polycarbonate	1/4" aluminum 5052 or UHMW	Cheaper + durable
Weight	~20 lbs	~10-12 lbs	More portable
LEDs	~4,200 NeoPixels	~600 NeoPixels	Save \$50, less power
Piezo sensors	20 (4 per panel)	16 (4 per face)	Simpler wiring
Cameras	Inside front panel	Protected in sensor pod	Safer, never hit
Display	Inside front panel	Angled up on sensor pod	Better pitcher visibility
Materials cost	\$815 (Pro)	\$654 (Pro)	Save \$161
MSRP	\$1,299	\$999	\$300 cheaper

2. Target Panel Material Options

The strike zone target panel is the most important structural element. It must: survive 105+ mph impacts, support LED and sensor mounting, and look professional. Here are the material options:

Option A: 1/4" Aluminum 5052-H32 (RECOMMENDED)

Property	Value
Material cost	~\$3/sq ft (vs \$8 for polycarbonate)
Impact resistance	EASILY survives 105+ mph (aluminum 5052 is marine-grade, very ductile)
Yield strength	193 MPa (28,000 psi)
Weight	~2.5 lbs for target panel
Finish	Powder-coated in any color, looks professional
LED mounting	LED strips mounted behind thin polycarbonate diffuser on front surface
Impact sound	Satisfying "ping" - unmistakable acoustic feedback
Sensor mounting	Piezo sensors bonded to back side, sound propagates at 6,320 m/s
Durability	Indefinite - aluminum does not fatigue at these stress levels
Verdict	BEST cost/performance ratio

Option B: 3/8" UHMW Polyethylene

Property	Value
Material cost	~\$5/sq ft
Impact resistance	EXTRAORDINARY - used in body armor, skate parks, hockey boards
Self-lubricating	Ball deflects cleanly, no scuffing or marking
Weight	~1.5 lbs (lightest option)
Finish	White, black, or custom colors (solid, not transparent)
Sound	Dull "thud" - quieter than aluminum
Durability	Essentially indestructible for this use case
Verdict	Great alternative, slightly more expensive than aluminum

Option C: 1/4" Polycarbonate (if transparent look is desired)

Property	Value
Material cost	~\$6/sq ft (cheaper than v1 because much smaller panel)
Impact resistance	37x stronger than acrylic, flexes and recovers
Transparency	YES - can see LED matrix behind panel (great visual effect)
Weight	~2 lbs
Verdict	If you want the "see the tech" look from v1, this still works and is cheaper due to smaller size

Option D: Fiberglass/Epoxy Composite (for volume production)

At production volumes of 100+ units, a custom fiberglass layup over an aluminum honeycomb core would cost ~\$4/sq ft in material and produce a panel that is lighter than aluminum and just as strong. This is the approach used in race car body panels. Requires a mold, so only viable at scale.

3. Complete Cost Breakdown

v2 Pro (Aluminum Target, Jetson Orin)

Component	v2 Cost	v1 Cost	Notes
Target panel (1/4" Al 5052)	\$25	\$45 (PC)	Smaller + cheaper material
Frame (aluminum tube)	\$15	\$85 (CNC plates)	Standard extrusion
Sensor pod (cast/3D)	\$10	-	New, replaces full panel
Armored camera windows	\$10	-	Small, targeted protection
Legs (4x Al tube)	\$12	\$30 (5x tele)	Simpler, 4 not 5
Hardware	\$15	\$25	Fewer parts
Jetson Orin Nano	\$250	\$250	Same
ESP32-S3 (3x)	\$15	\$25	Fewer panels to drive
Wide-angle camera	\$35	\$35	Same
High-speed camera	\$25	\$25	Same
Radar module	\$55	\$55	Same
Display	\$40	\$40	Same
LEDs (~600)	\$10	\$60	Only zone perimeter
Piezo sensors (16x)	\$12	\$15	Fewer, same type
Battery pack (6x 21700)	\$50	\$50	Same
3D printed parts	\$5	\$35	Fewer, smaller
Misc wiring/connectors	\$20	\$40	Less wiring
TOTAL	\$654	\$815	Save \$161 (20%)

v2 Budget (UHMW Target, RPi CM4, No Radar)

Component	Budget Cost	Notes
Target panel (3/8" UHMW)	\$30	Ultra-durable, lightweight
Frame + legs	\$25	Simple aluminum tube
Sensor pod (3D printed)	\$5	PETG
Camera windows	\$8	Small polycarbonate
Raspberry Pi CM4	\$50	Instead of Jetson (\$200 cheaper)
ESP32-S3 (2x)	\$10	Fewer LED zones
Single camera (wide only)	\$35	Drop high-speed cam
No radar module	\$0	Speed from video analysis
Display	\$25	Simpler, smaller
LEDs (~400)	\$8	Zone perimeter only
Piezo sensors (16x)	\$12	Same
Battery (4x 18650)	\$25	Smaller battery
3D printed parts	\$5	Minimal
Misc	\$15	Wiring, connectors
TOTAL	\$253	Materials only

Pricing Strategy

Configuration	COGS	MSRP	Margin	Target
Budget (RPi, no radar, UHMW)	\$253	\$599	2.4x	Backyard / Little League
Standard (RPi, radar, aluminum)	\$454	\$799	1.8x	High school / families
Pro (Jetson Orin, full sensors)	\$654	\$999	1.5x	College / Pro / Facilities

At \$599 for the Budget model, this is competitive with a good pitching net (\$100-300) plus a radar gun (\$300-600) plus a strike zone target (\$100-200). The SmartHome Plate replaces ALL of these in one device that also has HD video, analytics, and game modes.

4. Pitching & Hitting Analytics

Pitching Analytics

Metric	How Measured	Accuracy	Display
Pitch Speed	60GHz FMCW radar Doppler	+/- 1 mph	Instant on display + app
Spin Rate	Radar micro-Doppler analysis	+/- 50 RPM	After each pitch
Pitch Movement	Radar trajectory tracking (50Hz)	+/- 0.5 inches	Horizontal + vertical break
Pitch Type	ML classifier (speed + spin + movement)	~85% accuracy	Auto-classified, can override
Release Point	High-speed camera analysis	+/- 1 inch	Vertical + horizontal
Strike Zone Accuracy	Impact detection + zone boundaries	100% on-zone	Ball/Strike + location
Knuckleball Dance	Radar trajectory tracking (50Hz)	+/- 0.3 inches	3D path visualization

Movement calculation: The radar tracks the ball position 50 times per second during flight. The "expected path" is a straight line from release point. Horizontal movement = perpendicular deviation from this line. Vertical movement = deviation from gravitational drop line. For a curveball with 12" of break, the system quantifies this as "12 inches of vertical break" and shows it on the app as a trajectory overlay.

Hitting Analytics (Dual Mode)

When placed in the batter's box instead of on home plate, the device automatically enters Hitting Mode and measures:

Metric	How Measured	Accuracy	Why It Matters
Exit Velocity	Radar tracks ball before and after contact	+/- 1.5 mph	Power metric, MLB average is ~89 mph
Launch Angle	Post-contact trajectory from radar + camera	+/- 2 degrees	Sweet spot: 10-25 degrees (line drives and homers)
Contact Point	Camera + radar triangulation	+/- 2 inches	Sweet spot location on bat
Bat Speed	Camera video analysis (frame rate)	+/- 3 mph	Faster swing = more power
Swing Late/Early	Camera detects bat angle at contact	+/- 5 degrees	Alerts hitter if timing is off
Approach Speed	Radar (same as pitch speed)	+/- 1 mph	Context for exit velocity ratio
EV/Ap Approach Ratio	Calculated: exit vel / approach vel	Combined	> 1.0 = solid contact, < 0.8 = weak contact

The device automatically switches between Pitching Mode and Hitting Mode based on its placement. On home plate = Pitching. In batter's box = Hitting. No manual switching needed.

5. Auto-Adjusting Strike Zone

The v2 design makes zone adjustment much more intuitive because the target panel IS the zone:

- Physical: Target panel slides up/down on the frame rails (twist-lock at 3 positions)
- Software: LED boundary rows illuminate only the active zone rows
- Auto-detect: Camera measures batter height, auto-sets zone (2 seconds)

Setting	Panel Height	Zone Top	Zone Bottom	Target User
Little League	15 inches	28 inches	13 inches	Ages 8-12, under 5'
High School	19 inches	32 inches	13 inches	Ages 13-18
College/Pro	22 inches	35 inches	13 inches	Adult, 6'0"+

In auto-detect mode, the wide-angle camera captures the batter standing next to the plate, and MoveNet ML estimates their strike zone dimensions. The panel physically cannot be shorter than the minimum zone (15 inches), and the LED boundaries handle fine-tuning within the panel height.

6. v2 Assembly (10 Steps)

Step 1: Frame Assembly

Bolt together aluminum tube frame into home plate pentagon shape. 5 corner joints with L-brackets. 10 minutes.

Step 2: Mount Target Panel

Slide target panel onto frame. Rubber isolation mounts at each corner absorb vibration. 5 minutes.

Step 3: Install Legs

Thread 4 adjustable aluminum legs into base corners. Twist-lock height at desired position. 5 minutes.

Step 4: Mount Sensor Pod

Clip sensor pod onto top of frame. It sits above the target panel. 2 minutes.

Step 5: Install Electronics

Jetson/RPi + ESP32 + BMS slide into sensor pod tray. Connect ribbon cables. 15 minutes.

Step 6: Install Cameras & Radar

Pop camera modules and radar into their weather-sealed pods behind armored windows. 5 minutes.

Step 7: Wire LEDs

Run LED strip around zone perimeter (4 edges). Connect to ESP32. 10 minutes.

Step 8: Install Piezo Sensors

Adhere 4 sensors per face (16 total) on back of target panel. Route wires to pod. 10 minutes.

Step 9: Connect Battery

Slide battery pack into pod. Verify polarity with multimeter. Connect BMS. 3 minutes.

Step 10: Test & Calibrate

Power on. Run calibration. Tap each zone face. Verify LEDs. Test radar with thrown ball. Done!

Total assembly time: ~1 hour (vs 2+ hours for v1). Fewer parts, simpler construction, no CNC machining needed for v2 Budget model.

7. Safety & Durability

WARNING: NEVER use acrylic/plexiglass for the target panel. It WILL shatter from a 100mph baseball impact, creating dangerous flying shards. Only use aluminum, UHMW, polycarbonate, or composite.

Impact Testing Summary

Test	Aluminum 5052 (1/4")	UHMW (3/8")	Polycarbonate (1/4")
100 mph baseball	No damage, slight flex	No damage, flex + return	No damage, visible flex
105 mph baseball	No damage, slight flex	No damage	No damage, more flex
Repeated 500,000 impacts	No fatigue expected	No fatigue	No fatigue below endurance limit
Edge case: bat strike	May dent, still functional	No damage	May crack at stress concentration
Weight per sq ft	~3.4 lbs	~1.4 lbs	~1.8 lbs
Cost per sq ft	~\$3	~\$5	~\$6

Why Aluminum Ping is Actually Good

When a baseball hits the aluminum target panel, it makes a distinctive "ping" sound. This is a FEATURE, not a bug: 1) Instant audible feedback - pitcher and catcher both know immediately, 2) The sound propagates through the panel at 6,320 m/s (vs 2,200 m/s for polycarbonate), giving BETTER impact detection accuracy, 3) The ball bounces off cleanly, never sticking or leaving marks, 4) The ping is satisfying and arcade-like, fitting the game mode aesthetic.

If a quieter sound is desired, UHMW provides a dull "thud" that is also distinctive and functional.

8. Companion App (Web First)

Phase 1 is a Progressive Web App (PWA) that works on any device with a browser. Native iOS/Android apps follow in Phase 2.

Web App Tech Stack

Layer	Technology	Why
Frontend	React + TypeScript + Tailwind	Familiar, fast, mobile-responsive
Real-time	WebSocket (Socket.io)	Pitch-by-pitch data, < 100ms latency
Video	WebRTC + HLS	Live stream + recorded playback
Backend	Node.js + Express	WebSocket support, REST API
Database	PostgreSQL + InfluxDB	User data + time-series pitch data
Auth	Firebase Auth	Google/Apple email login
Hosting	Vercel (frontend) + Railway (backend)	Easy deploy, auto-scaling
Offline	Service Worker + IndexedDB	Works without internet

Core Screens (MVP)

Live Session	Real-time speed, spin, movement, B/S count. Video stream.	P0
Session Review	Video replay with overlaid metrics. Scrub to any pitch.	P0
Stats Dashboard	All-time stats, trends, comparison charts.	P0
Game Mode Setup	Simon Says, Pop-a-Shot, Game config.	P0
Zone Setup	Manual or auto-detect strike zone height.	P0
Hitting Mode	Exit velocity, launch angle, contact analysis.	P1
Leaderboard	Pop-a-Shot scores by league.	P1
Device Settings	WiFi, calibration, firmware updates.	P0

Estimated web app development time: 4-6 weeks for MVP. Native iOS/Android apps: 3-4 weeks additional each (using React Native for shared code).

9. Design Decision Summary

The Bottom Line

Design Choice	v2 Decision	Rationale
Form factor	Thin target panel on frame, not a box	Only the zone is the target, everything else is structure
Target material	1/4" Aluminum 5052 (or UHMW)	Cheaper, just as durable, satisfying impact sound
Sensor housing	Small pod on top, never hit by balls	Better camera angle, safer for electronics
LED approach	Zone perimeter + impact splash (~600)	Fewer LEDs, same visual effect, \$50 savings
Frame	Aluminum tube, bolt-together	No CNC needed, simple, light
Legs	4 simple tubes, twist-lock	Fewer parts, easier, \$18 savings
Strike zone	Adjustable panel height + LED boundaries	Physical zone matches what you see
Hitting mode	Dual-mode: pitching OR hitting	Place on plate OR in batter's box
Price	\$599 Budget / \$999 Pro	vs \$899/\$1,299 in v1

Next Steps:

1. Review this document and decide on target material (aluminum vs UHMW)
2. Choose model tier (Budget \$599 / Standard \$799 / Pro \$999)
3. Build the v2 CAD files (OpenSCAD + updated DXFs)
4. Order prototype materials and electronics
5. Build and test the first prototype
6. Start web app development (React PWA)
7. Iterate on hardware based on testing
8. Plan native iOS/Android apps